AOC-IDS: Autonomous Online Framework with Contrastive Learning for Intrusion Detection

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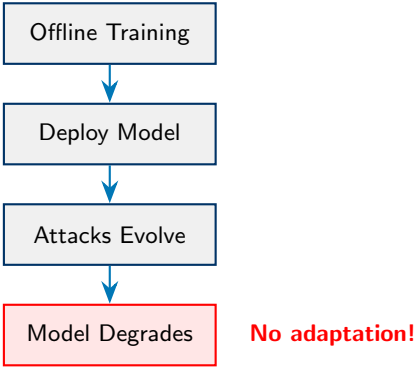
Outline

1. Background & Motivation
2. AOC-IDS Framework
3. Our Novel Contributions
4. Summary

The Problem with Traditional IDS

Challenges in real-world intrusion detection:

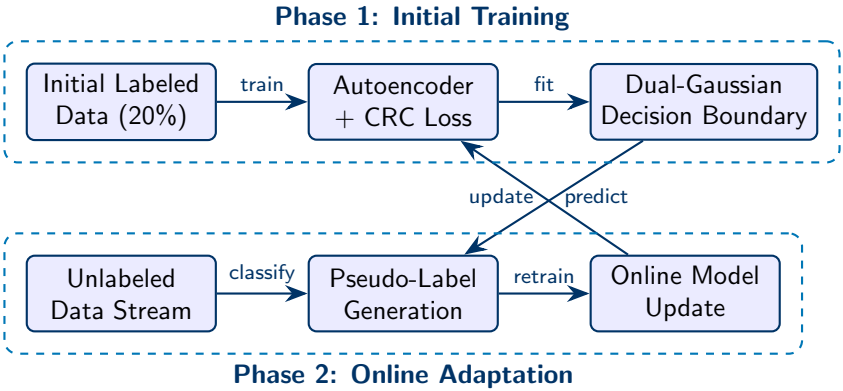
- Traditional IDS relies on **offline, static training**
- Attack strategies **evolve continuously**
- Manual labeling of network traffic is **expensive** and **slow**
- IoT environments generate **massive, dynamic** data streams
- Models degrade over time without retraining (**concept drift**)



Key Question

How can an IDS *autonomously* adapt to new traffic patterns *without* human labeling?

AOC-IDS: System Overview



Core Component: Autoencoder + CRC Loss

Autoencoder Architecture

- Encoder: $d \to d/2 \to d/4$ (bottleneck)
- Decoder: $d/4 \to d/2 \to d$ (reconstruction)
- Outputs **two** representation spaces:
  - Encoder embeddings (compressed features)
  - Decoder reconstructions

Total Loss:

$$\mathcal{L} = \mathcal{L}_{\text{CRC}}^{\text{encoder}} + \mathcal{L}_{\text{CRC}}^{\text{decoder}}$$

Cluster Repelling Contrastive (CRC) Loss

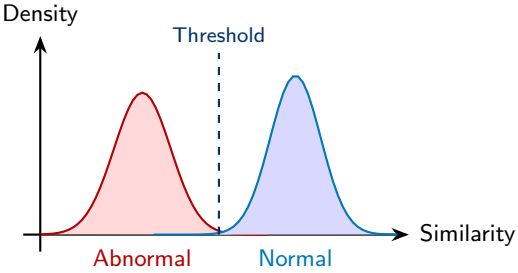
- Built on InfoNCE, but tailored for anomaly detection.
- **Crucial difference**: It incorporates **two traversals** of the abnormal set, amplifying the repelling force.
- **Intuition**: Forces normal traffic into a tight cluster while pushing known anomalies far away in the embedding space.

$$\mathcal{L}_{\text{CRC}} = -\frac{1}{N_n} \sum \log \frac{e^{s_{ij}/\tau}}{e^{s_{ij}/\tau} + \sum e^{s_{ik}/\tau}}$$

Detection: Dual-Gaussian Decision Boundary

How detection works:

1. Compute **normal template** = mean embedding of all normal training samples
2. For each sample, compute **cosine similarity** to normal template
3. Fit **two Gaussians** (via MLE) to the similarity distribution:
   - $\mathcal{N}(\mu_1, \sigma_1)$ — normal cluster (high similarity)
   - $\mathcal{N}(\mu_2, \sigma_2)$ — abnormal cluster (low similarity)
4. Classify: if $p(\text{abnormal} \mid x) > p(\text{normal} \mid x)$ $\Rightarrow$ **attack**

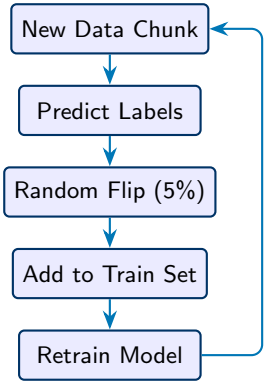


Final vote: encoder and decoder predictions combined by confidence ($|p_1 - p_2|$).

Online Pseudo-Labeling Loop

How online adaptation works:

1. Start with a small labeled set (20% of training data)
2. Train the AE + CRC model on this initial set
3. Receive a **chunk** of unlabeled data (size = `sample_interval`)
4. Use Gaussian decision boundary to **predict pseudo-labels**
5. **Flip** a small fraction of labels (5%) for robustness
6. Add pseudo-labeled data to training set
7. **Retrain** model for `epoch_1` epochs
8. Repeat until all data is consumed



Experimental Setup (Original Paper)

Datasets:

- **NSL-KDD** — 121 features, binary labels
- **UNSW-NB15** — 196 features, binary labels
- Preprocessed: normalized continuous, one-hot categorical

Setup:

- 20% initial labeled data
- 80% arrives as unlabeled stream
- 5 random seeds, averaged results
- Evaluated: Accuracy, Precision, Recall, F1

Key findings from the paper:

- AOC-IDS **outperforms** state-of-the-art on both datasets
- Achieves strong results with **only 20%** labeled data
- Online adaptation **improves** performance over the static baseline
- CRC loss is **more effective** than standard contrastive losses (InfoNCE)

Significance

First framework combining contrastive learning with autonomous online adaptation for IDS.

Evaluation: Reproduction & Novelty Results

Table: **NSL-KDD Dataset (%)**

Model	Accuracy	Precision	Recall	F1 Score
Original Paper Reported	88.90	85.99	96.21	90.81
Our Baseline Reproduction	86.91	83.28	96.35	89.34
Our HNSW Novelty	TBD	TBD	TBD	TBD

Table: **UNSW-NB15 Dataset (%)**

Model	Accuracy	Precision	Recall	F1 Score
Original Paper Reported	89.19	90.65	89.70	90.14
Our Baseline Reproduction	89.41	91.30	89.26	90.27
Our HNSW Novelty	87.00	88.30	89.04	89.11

Evolution of the Decision Boundary: What Didn't Work

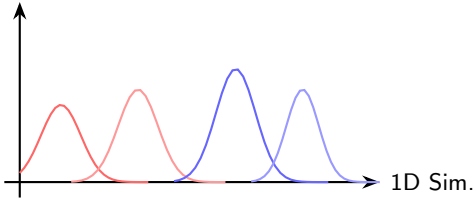
Initial Hypothesis: GMM

- The original paper forces cosine similarities into a **1D 2-Gaussian** distribution.
- **Idea:** Use a Gaussian Mixture Model (GMM) with BIC selection to capture multiple attack types.

Why it failed:

- Still projects high-dimensional embeddings into a **1D similarity space**, losing crucial structural information.
- Optimization was **unstable** online (Nelder-Mead/EM algorithms struggled with continuous data streams).

Failed Attempt: GMM in 1D



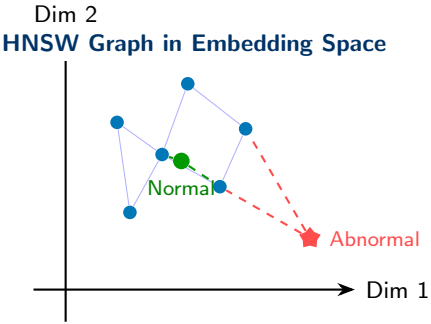
The Realization

The problem isn't the distribution shape; it's the **lossy 1D projection**.

Novelty 1: High-Dimensional HNSW Vector Search

The Solution: Stay in High-D Space

- Instead of projecting to 1D, we use the **full 32/64-dimensional autoencoder embeddings**.
- **Approach:** Build a Hierarchical Navigable Small World (**HNSW**) graph of *only normal* training samples.
- **Detection:** For a test sample, query the k-Nearest Neighbors in the HNSW graph.
- If the mean k-NN distance $>$ threshold (95th percentile) $\Rightarrow$ **Anomaly**.



Advantages:

- Captures highly **non-linear** normal clusters.
- **Fast inference** ($O(\log N)$) suitable for

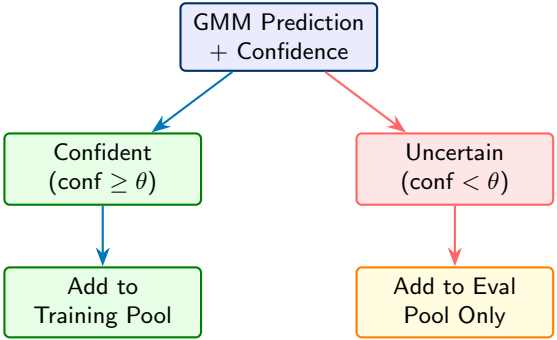
Novelty 2: Confidence-Aware Pseudo-Label Filtering

Problem with original approach:

- **All** pseudo-labels added to training — including wrong ones
- Random flipping (5%) is **blind**: flips correct labels too
- Noisy labels **accumulate** over online rounds
- Model performance degrades with error propagation

Our improvement:

- Compute **confidence** = $|P(\text{normal}) - P(\text{abnormal})|$ from GMM posteriors
- Only add samples with confidence $> \theta$



Result: Cleaner training data $\Rightarrow$ better model $\Rightarrow$ better future predictions (virtuous cycle)

Novelty 3: Dynamic Temperature Scheduling

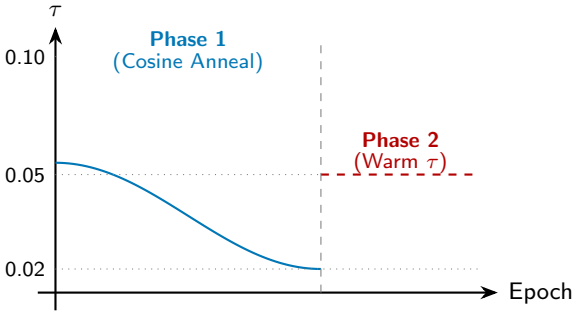
Problem with original approach:

- Fixed temperature $\tau = 0.02$ throughout training
- Too aggressive early on (embeddings still forming)
- Same τ for initial and online phases, but they have **different needs**

Our improvement — Cosine Annealing:

$$\tau(t) = \tau_{\min} + \frac{\tau_{\max} - \tau_{\min}}{2}\left(1 + \cos\frac{\pi\, t}{T}\right)$$

- **Phase 1** (initial): τ anneals from $0.1 \rightarrow 0.02$



High τ = soft separation (exploration)

Low τ = hard separation (exploitation)

Summary of Contributions

Aspect	Original AOC-IDS	Our Improvement
Decision Boundary	1D 2-Gaussian MLE	**High-D HNSW Vector Search**
Pseudo-Label Quality	Random flip (5%)	Confidence-based filtering
CRC Temperature	Fixed $\tau = 0.02$	Cosine annealing + warm online τ

Why These Improvements Matter

- **HNSW Vector Search**: Replaces lossy 1D assumption with true spatial geometry $\Rightarrow$ better detection of complex anomalies.
- **Confidence filtering**: Breaks error propagation cycle $\Rightarrow$ cleaner online learning.
- **Temperature scheduling**: Matches training dynamics $\Rightarrow$ stable convergence.

Thank You!